

A top-down view of a light teal desk. In the top left is a silver laptop with a black keyboard. To its right is a white mouse and a dark coffee cup. In the bottom left is a brown leather notebook with a white pen and a pair of tortoiseshell glasses.

AIClub





Basics of Data Literacy

What is Data?

- **Data = Facts or information collected for analysis.**

- Examples:

- Marks in exams
- Daily temperature
- Number of visitors
- Survey responses

- Data can be:

- Numbers
- Text
- Images
- Audio



Numbers



Text



Images

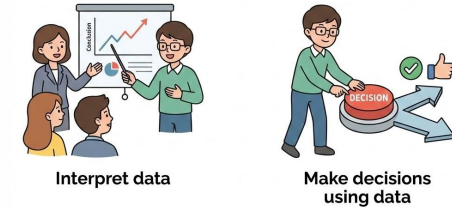
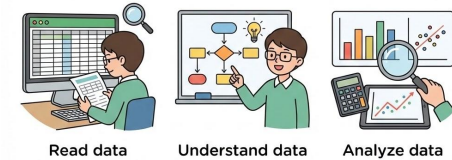


Audio

Meaning of Data Literacy

Data Literacy is the ability to:

- Read data
- Understand data
- Analyze data
- Interpret data
- Make decisions using data

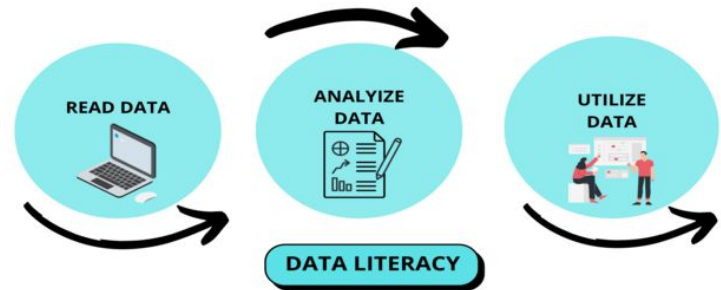


In simple words: 'Using data smartly and responsibly.'

Why is Data Literacy Important?

It helps us to:

- Make informed decisions
- Avoid misinformation
- Think critically
- Identify trends
- Solve real-world problems



Example: School checks result data before planning remedial classes.

What are Data Points?

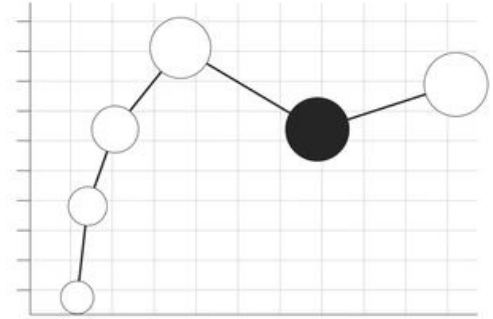
A Data Point is a single piece of information.

Example:

- One student's marks = one data point
- One temperature reading = one data point

Multiple data points together form a dataset.

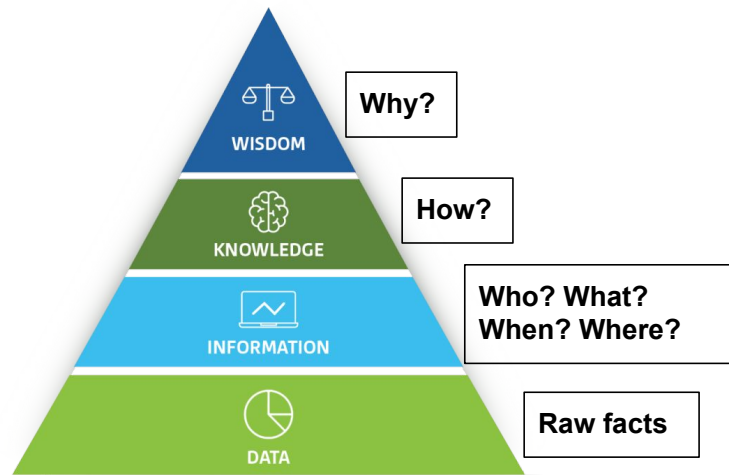
Example: Marks of 40 students = dataset.



The Data Pyramid

Data Pyramid shows how raw facts become knowledge.

- Wisdom
- Knowledge
- Information
- Data (Base)



Flow: Raw Data → Organized → Meaningful Information → Knowledge → Smart Decisions.

Data-Driven Decision Making

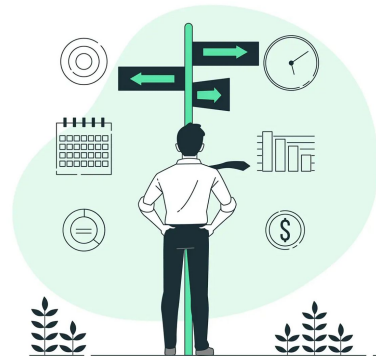
Decision based on facts, not guesswork.

Example:

❌ 'I think students struggle in Math.'

✓ 'Math average is 45% — extra classes needed.'

Data → Pattern → Insight → Action.



Impact of Data Literacy – Individual Level

Data literacy helps individuals:

- Make smart financial decisions
- Identify fake news
- Protect personal data
- Improve academic performance
- Make informed career choices



Impact of Data Literacy – Societal Level

Data literacy helps society:

- Improve healthcare systems
- Better governance decisions
- Disaster management
- Economic planning
- Smart city development



Data-aware citizens = Stronger society.

How to Become Data Literate?

- 1 Ask questions about data
- 2 Check data sources
- 3 Learn basic statistics (mean, median)
- 4 Practice reading graphs
- 5 Use spreadsheets
- 6 Think critically before believing data
- 7 Protect personal information.



Practice + Awareness = Data Literacy.

Data Privacy

Data Privacy = Who can access personal data?

Examples:

- Name
- Phone number
- Aadhaar number
- Password



Focus: Rights and permissions.

Data Security

Data Security = How data is protected from misuse.

Protection methods:

- Strong passwords
- Encryption
- Antivirus
- Firewalls.

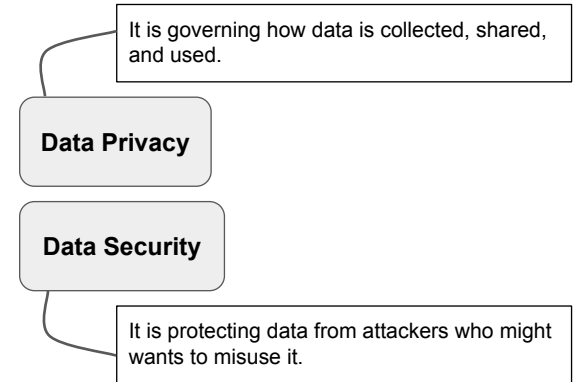
Focus: Protection systems.



Data Privacy vs Data Security

Data Privacy	Data Security
Who can access	How it is protected
Personal control	Technical protection
Focus on rights	Focus on safety

Both are essential in AI systems.



Cyber Safety Practices

Do's:

- ✓ Use strong passwords
- ✓ Don't share OTPs
- ✓ Think before clicking links
- ✓ Avoid public Wi-Fi for banking
- ✓ Update software regularly
- ✓ Report suspicious activity
- ✓ Activate Two-Factor Authentication (2FA)
- ✓ Download software from trusted sources



Cyber Safety Practices

Do's:

- ✓ Prioritize websites with "https://" for secure logins.
- ✓ Keep your browser, OS, and antivirus updated regularly.
- ✓ Adjust social media privacy for limited visibility
- ✓ Always lock your screen when away.
- ✓ Connect only with trusted individuals online.
- ✓ Report online bullying to a trusted adult immediately.
- ✓ Limit Permissions on Apps

Cyber Safety Practices

Don't's:

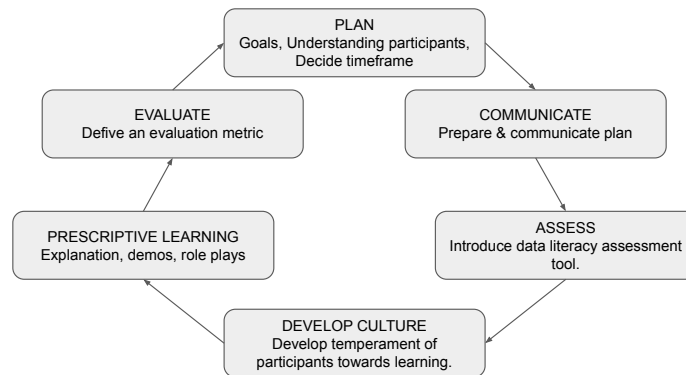
- ✖ Avoid sharing personal info like real name or phone number
- ✖ Don't send pictures to strangers or post them on social media
- ✖ Don't open emails or attachments from unknown sources
- ✖ Ignore suspicious requests for personal info like bank account details
- ✖ Keep passwords and security questions private
- ✖ Don't copy copyrighted software without permission
- ✖ Avoid cyberbullying or using offensive language online



Data Literacy Process Framework

Data Literacy Process involves:

- 1 Ask Questions
- 2 Collect Data
- 3 Clean & Organize Data
- 4 Analyze Data
- 5 Interpret Results
- 6 Communicate Insights
- 7 Take Action.



Data → Understanding → Decision.

Case Study – School Attendance

Situation: School wants to improve attendance.

Step 1: Collect attendance data

Step 2: Identify pattern (low attendance on Mondays)

Step 3: Investigate reasons

Step 4: Take action (engaging Monday activities).

Result: Improved attendance.



Case Study – Fake News Detection

Situation:

A viral message claims a holiday announcement.

- Data Literate Student:
- Checks official website
- Verified source
- Confirms authenticity

Data literacy prevents misinformation.



Quick Recap

- ✓ Data points form datasets
- ✓ Data Pyramid shows data transformation
- ✓ Data-driven decisions use facts
- ✓ Privacy $\neq$ Security
- ✓ Data literacy impacts individuals & society.

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THANK YOU



info@pyxeda.ai



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